

This report documents the trajectory simulations submitted under Section 10 of the LASC Rocket Challenge Standards Manual (RCSM Ed. 7 Rev. 1). It presents the **nominal** and **ballistic** flight cases required by CRS 10.1.11, the aerodynamic and mass-properties data behind them, an independent analytical verification of the stability solution, and a correlation of the model against measured flight-test telemetry.

|                                                                                |                                                                               |                                                                                      |                                                                                        |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <div>APOGEE (NOMINAL)</div> <div>1052.1 m</div> <div>AGL, at t = 15.91 s</div> | <div>MAX VELOCITY</div> <div>155.3 m/s</div> <div>Mach 0.463 at burnout</div> | <div>STATIC MARGIN</div> <div>1.92 - 2.57 cal</div> <div>entire powered ascent</div> | <div>MODEL ACCURACY</div> <div>0.71 %</div> <div>apogee error, 4 Jul 2026 flight</div> |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

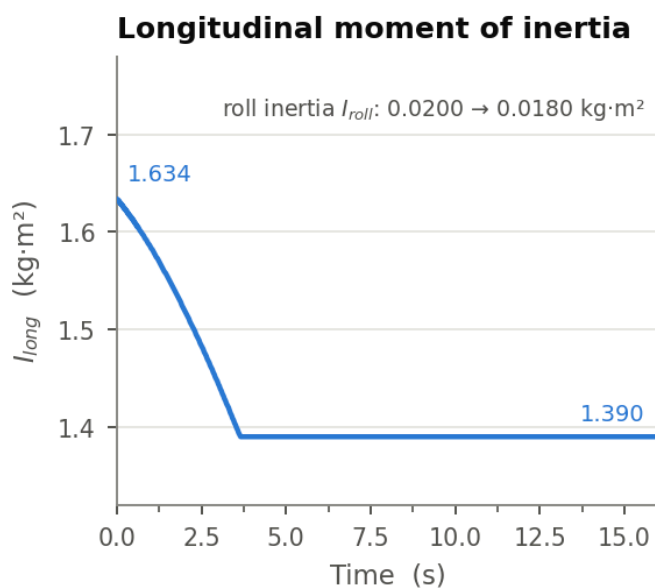
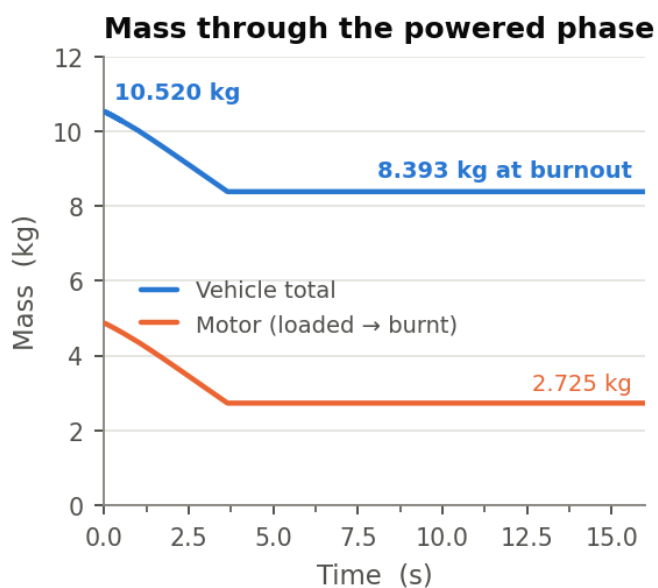
## 1 DELIVERABLES AND SIMULATION SET-UP

| Item                      | Value / file                                                                       |
|---------------------------|------------------------------------------------------------------------------------|
| Simulation tool           | OpenRocket 24.12 (6-DOF, Barrowman + extended aerodynamics)                        |
| Vehicle model             | FinalPrometeo2026.ork — component-level model, measured masses                     |
| Motor thrust curve        | Icarus-I_K503.eng — SRAD motor, supplied per CRS 10.1.9                            |
| Motor internal ballistics | OpenMotor — Lo que te llego.ric, KNSB / 2× BATES grain                             |
| Nominal flight case       | Simulation “brasil 2026” — parachute deployed at apogee (CRS 10.1.11)              |
| Ballistic flight case     | Simulation “Ballistic Brasil” — recovery inhibited (CRS 10.1.10, 10.1.11)          |
| Launch site               | Iacanga, SP — 21.900° S, 48.960° W, 490 m MSL                                      |
| Launch rail               | 4.0 m LASC-provided rail (FLT 4.4.1), elevation 80° (10° from vertical), into wind |
| Atmosphere                | ISA, 955.8 mbar / 11.8 °C at pad, $\rho = 1.168 \text{ kg/m}^3$                    |
| Wind                      | 2.7 m/s average, 10 % turbulence intensity                                         |
| Integration               | Time step 0.05 s, spherical geodetic model                                         |

## 2 VEHICLE CONFIGURATION AND MASS PROPERTIES

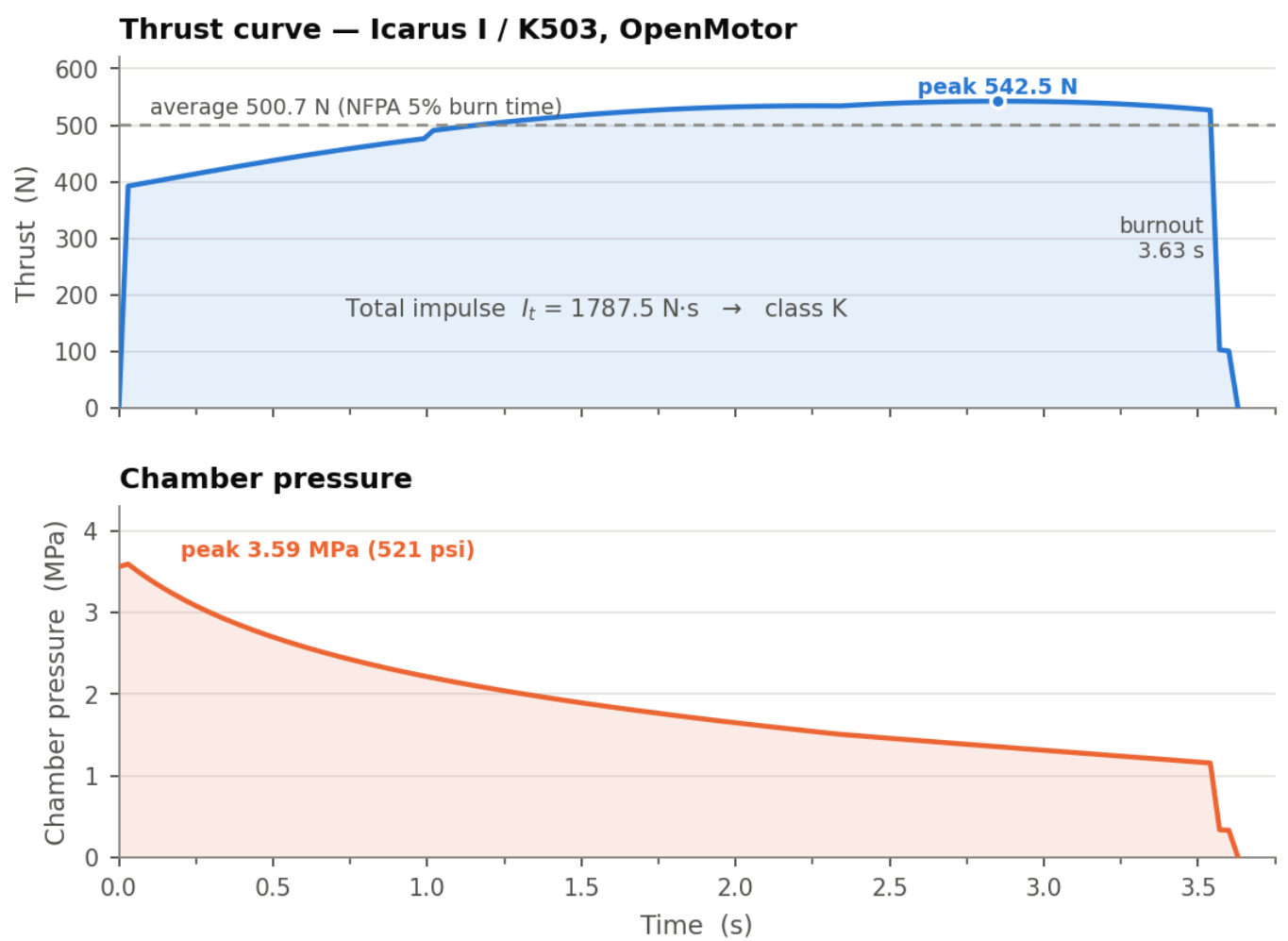
| Parameter                          | Value                                                       |
|------------------------------------|-------------------------------------------------------------|
| Overall length                     | 1.470 m                                                     |
| Airframe outer diameter            | 0.1096 m (reference calibre)                                |
| Reference area                     | 94.34 cm <sup>2</sup>                                       |
| Nose cone                          | Ellipsoidal, 0.270 m, acrylic, 4.0 mm wall                  |
| Body tube                          | 1.200 m, PVC, 4.5 mm wall                                   |
| Fins                               | 4 × trapezoidal, fibreglass, 3.0 mm, rounded LE/TE, 0° cant |
| Takeoff mass (model)               | 10.520 kg                                                   |
| Burnout mass                       | 8.393 kg                                                    |
| Propellant mass                    | 2.139 kg                                                    |
| CG at launch                       | 100.02 cm from nose tip                                     |
| CG at burnout                      | 92.88 cm from nose tip                                      |
| CP (Barrowman)                     | 121.03 cm from nose tip                                     |
| $I_{\text{long}}$ launch → burnout | 1.634 → 1.390 kg·m <sup>2</sup>                             |
| $I_{\text{roll}}$ launch → burnout | 0.0200 → 0.0180 kg·m <sup>2</sup>                           |

| Component                   | Mass (kg)     |
|-----------------------------|---------------|
| Motor, Icarus I (loaded)    | 4.864         |
| Fuselage, PVC               | 1.475         |
| Payload "PEZ"               | 1.000         |
| Parachute + harness         | 0.558         |
| Recovery system             | 0.517         |
| Ballast / mass component    | 0.506         |
| Fins (4, fibreglass)        | 0.505         |
| Modelling paste             | 0.500         |
| Nose cone, acrylic          | 0.227         |
| Bulkheads (3 × 75 g)        | 0.225         |
| Centring rings + inner tube | 0.143         |
| <b>Total at launch</b>      | <b>10.520</b> |



**Figure 1.** Mass and longitudinal inertia through the powered phase. The itemised budget at right above closes exactly on the 10.520 kg used by the trajectory solution.

3 PROPULSION — ICARUS I (K503)

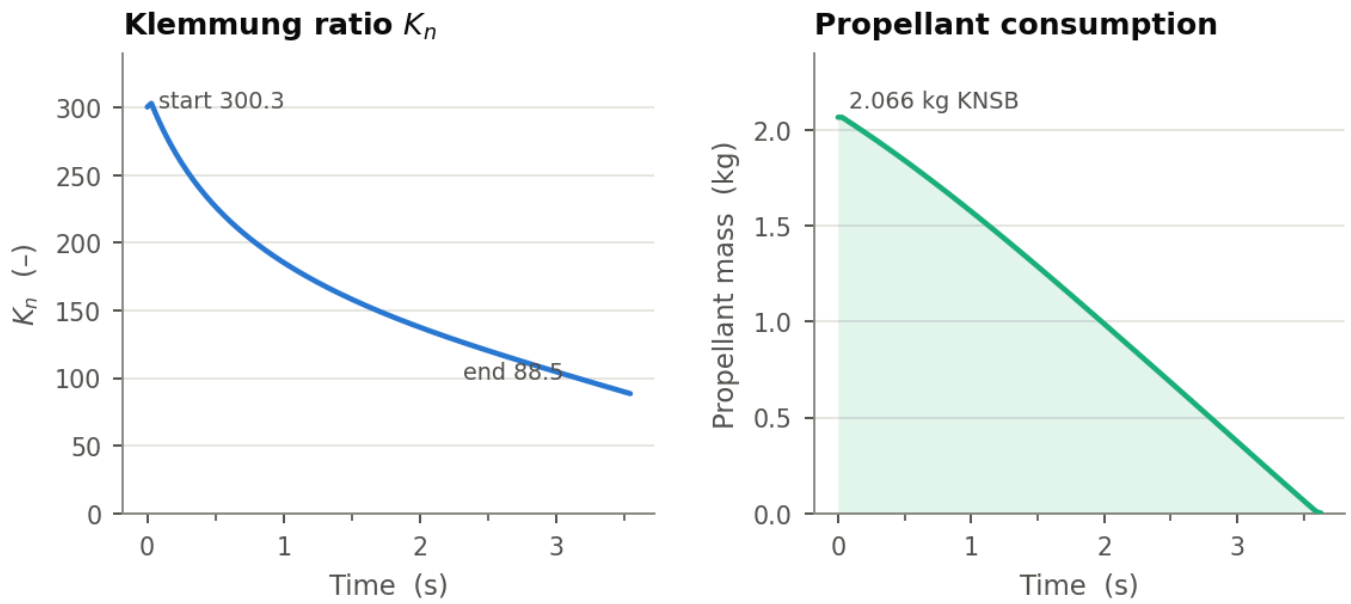


**Figure 2.** Icarus I thrust and chamber-pressure histories from the OpenMotor internal-ballistics solution (KNSB, two BATES grains, 11.90 mm throat,  $\epsilon = 9.6$ ).

| Parameter                      | Value                                                                |
|--------------------------------|----------------------------------------------------------------------|
| Propellant                     | KNSB (potassium nitrate / sorbitol),<br>$\rho = 1750 \text{ kg/m}^3$ |
| Grain configuration            | 2 × BATES, 74.93 mm OD, 18.54 /<br>19.05 mm core                     |
| Grain lengths                  | 134.6 mm + 151.1 mm                                                  |
| Nozzle throat / exit           | 11.90 mm / 36.87 mm — expansion<br>ratio 9.60                        |
| Nozzle half-angles             | 45° convergent, 15° divergent,<br>efficiency 0.70                    |
| Peak chamber pressure          | 3.59 MPa (521 psi)                                                   |
| Initial / final $K_n$          | 300.3 → 88.5                                                         |
| Propellant mass<br>(OpenMotor) | 2.067 kg                                                             |

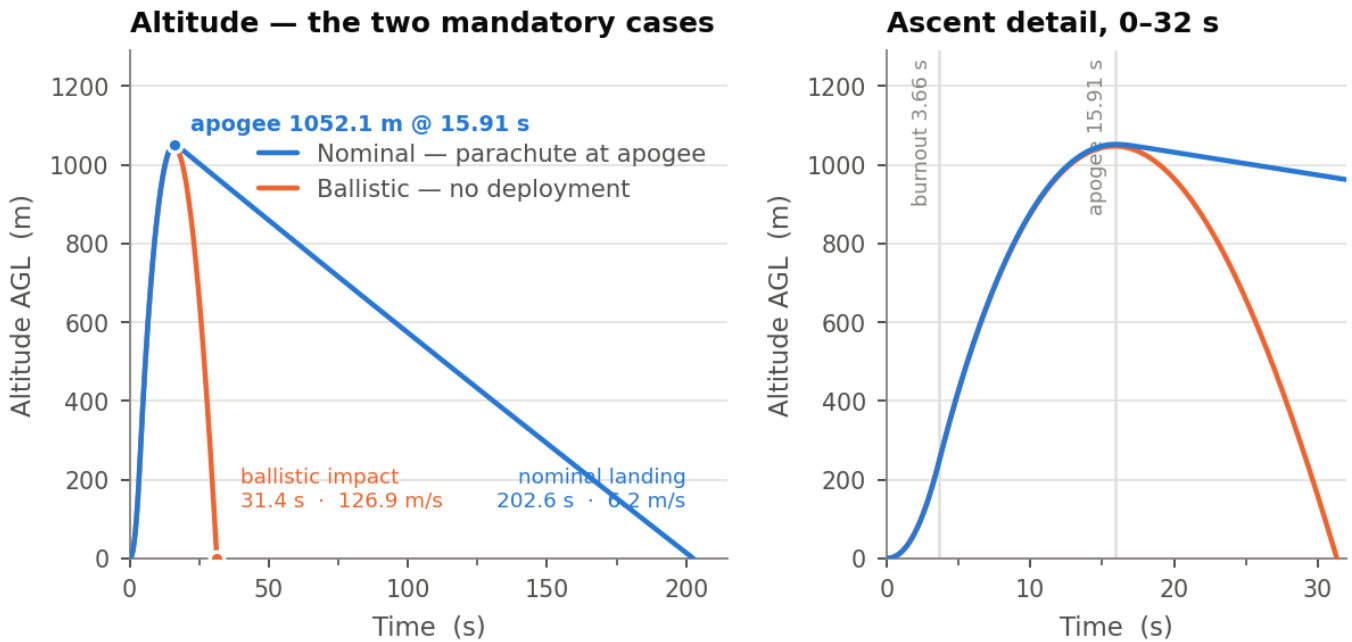
| Performance                        | OpenMotor  | Flight model<br>(.eng) |
|------------------------------------|------------|------------------------|
| Total impulse                      | 1787.5 N·s | 1855.9 N·s             |
| Average thrust (NFPA 5 %)          | 500.7 N    | 514.1 N                |
| Peak thrust                        | 542.5 N    | 551.0 N                |
| Burn time                          | 3.57 s     | 3.61 s                 |
| Motor class                        | K          | K                      |
| Total impulse limit (PRS<br>5.1.3) | 40 960 N·s | 4.5 % of limit         |

The `.eng` curve delivered with this submission is the one embedded in the OpenRocket model, so the trajectory results below are exactly reproducible from the delivered files. The OpenMotor solution is 3.7 % lower in total impulse; the difference is the nozzle-efficiency and erosion treatment. Both are reported for traceability.



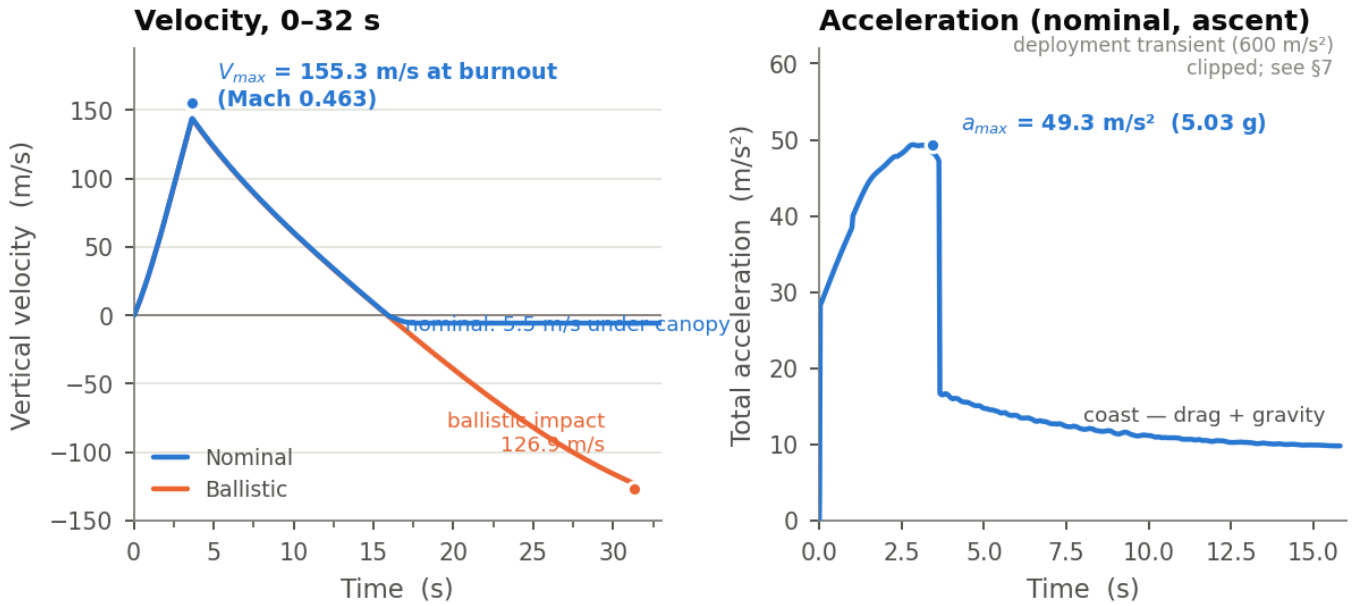
**Figure 3.** Klemmung ratio and propellant consumption.  $K_n$  falls monotonically through the burn — regressive, with no pressure spike.

## 4 TRAJECTORY — NOMINAL AND BALLISTIC CASES

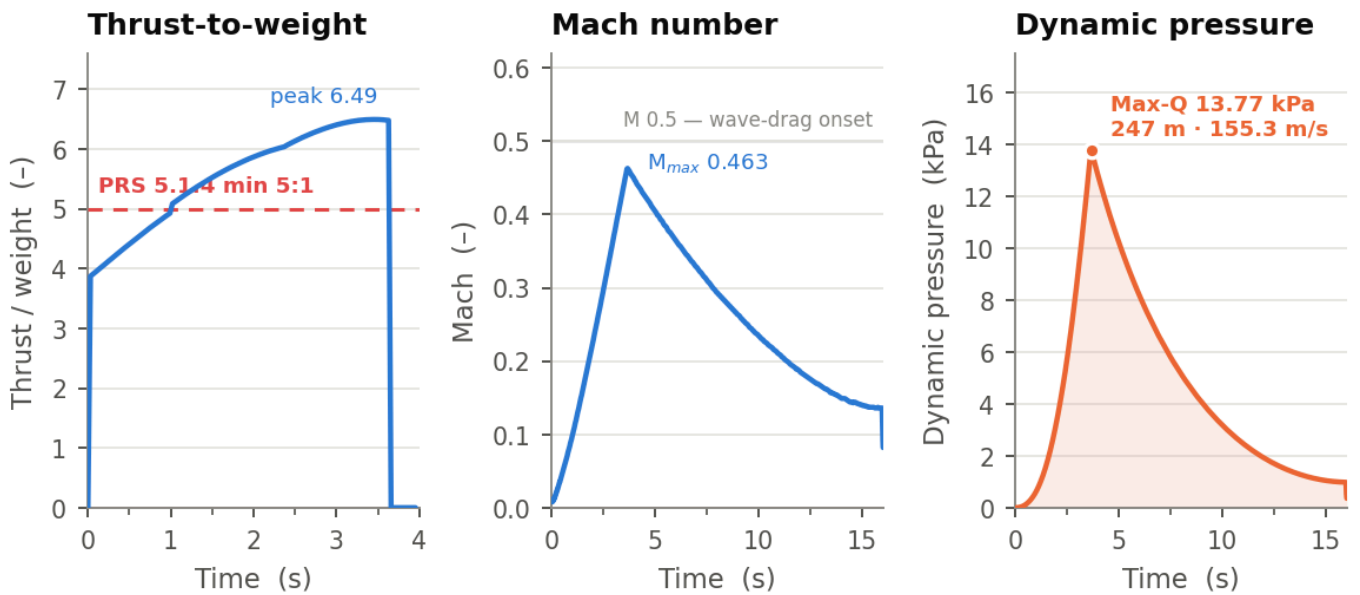


**Figure 4.** The two flight cases required by CRS 10.1.11. Both use identical vehicle, motor, atmosphere and wind; they differ only in whether the recovery device is allowed to deploy.

| Quantity                        | Nominal                         | Ballistic              |
|---------------------------------|---------------------------------|------------------------|
| Apogee, AGL                     | 1052.1 m                        | 1047.3 m               |
| Time to apogee                  | 15.91 s                         | 15.85 s                |
| Maximum velocity                | 155.33 m/s                      | 155.33 m/s             |
| Maximum Mach                    | 0.463                           | 0.464                  |
| Maximum acceleration            | 49.35 m/s <sup>2</sup> (5.03 g) | 49.39 m/s <sup>2</sup> |
| Rail departure velocity (4.0 m) | 16.17 m/s                       | 16.17 m/s              |
| Burnout time / altitude         | 3.66 s / 247.1 m                | 3.66 s / 247.1 m       |
| Max-Q                           | 13.77 kPa at 247 m              | 13.77 kPa              |
| Recovery deployment             | apogee, at 43.1 m/s             | —                      |
| Steady descent rate             | 5.52 m/s                        | —                      |
| Total flight time               | 202.6 s                         | 31.35 s                |
| Impact velocity                 | 6.17 m/s                        | 126.86 m/s             |
| Impact kinetic energy           | 0.16 kJ                         | 67.5 kJ                |
| Distance from pad at landing    | 193 m                           | 1278 m                 |



**Figure 5.** Velocity and acceleration. The two cases are indistinguishable up to apogee; they separate entirely on the descent.



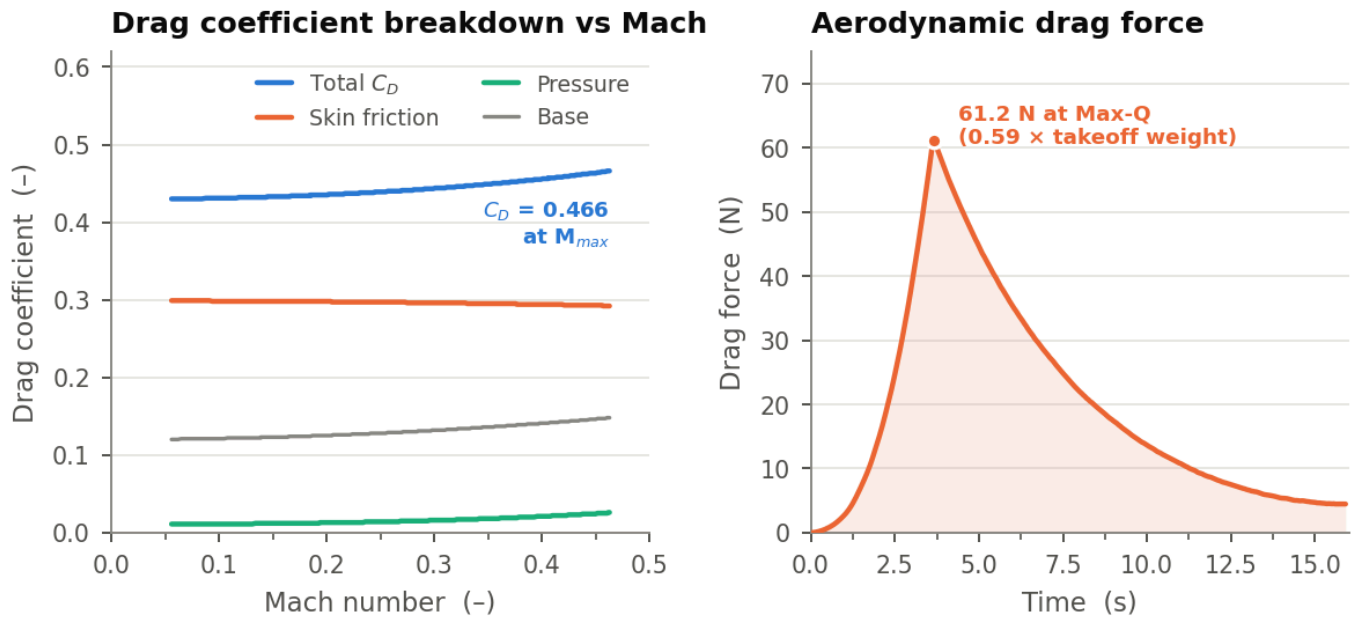
**Figure 6.** Thrust-to-weight, Mach number and dynamic pressure. Max-Q coincides with burnout — 13.77 kPa at 247 m AGL, well below any structural concern for this airframe.

#### 4.1 Trajectory data

| t (s) | Alt AGL (m) | V (m/s) | a (m/s²) | Mach   | q (kPa) | Thrust (N) | Mass (kg) | CG (cm) | SM (cal) | Drag (N) | $I_{long}$ (kg·m²) |
|-------|-------------|---------|----------|--------|---------|------------|-----------|---------|----------|----------|--------------------|
| 0.00  | 0.0         | 0.00    | 0.00     | 0.0080 | 0.00    | 0.0        | 10.520    | 100.02  | 1.917    | 0.0      | 1.6340             |
| 0.06  | 0.0         | 1.17    | 28.58    | 0.0090 | 0.00    | 401.4      | 10.500    | 99.97   | 1.922    | 0.0      | 1.6320             |
| 0.54  | 4.0         | 16.17   | 33.90    | 0.0500 | 0.15    | 447.6      | 10.270    | 99.33   | 1.980    | 0.8      | 1.6090             |
| 0.99  | 14.7        | 32.45   | 38.44    | 0.0980 | 0.61    | 483.1      | 10.030    | 98.63   | 2.044    | 2.7      | 1.5850             |
| 2.01  | 67.6        | 76.80   | 46.70    | 0.2290 | 3.42    | 540.9      | 9.411     | 96.71   | 2.219    | 14.5     | 1.5190             |
| 3.00  | 161.0       | 124.35  | 49.22    | 0.3710 | 8.90    | 550.6      | 8.788     | 94.49   | 2.421    | 38.5     | 1.4440             |
| 3.66  | 247.1       | 155.33  | 16.67    | 0.4630 | 13.77   | 0.0        | 8.384     | 92.88   | 2.569    | 61.2     | 1.3900             |
| 5.00  | 426.2       | 134.80  | 14.72    | 0.4040 | 10.18   | 0.0        | 8.384     | 92.88   | 2.569    | 44.6     | 1.3900             |
| 8.01  | 733.9       | 97.20   | 11.98    | 0.2930 | 5.14    | 0.0        | 8.384     | 92.88   | 2.569    | 21.9     | 1.3900             |
| 12.01 | 976.8       | 60.03   | 10.43    | 0.1850 | 1.91    | 0.0        | 8.384     | 92.88   | 2.569    | 8.4      | 1.3900             |

| t (s) | Alt AGL<br>(m) | V<br>(m/s) | a<br>(m/s <sup>2</sup> ) | Mach   | q<br>(kPa) | Thrust<br>(N) | Mass<br>(kg) | CG<br>(cm) | SM<br>(cal) | Drag<br>(N) | I <sub>long</sub><br>(kg·m <sup>2</sup> ) |
|-------|----------------|------------|--------------------------|--------|------------|---------------|--------------|------------|-------------|-------------|-------------------------------------------|
| 15.90 | 1052.1         | 43.11      | 9.79                     | 0.1360 | 0.98       | 0.0           | 8.384        | 92.88      | 2.569       | 4.5         | 1.3900                                    |

## 5 AERODYNAMICS

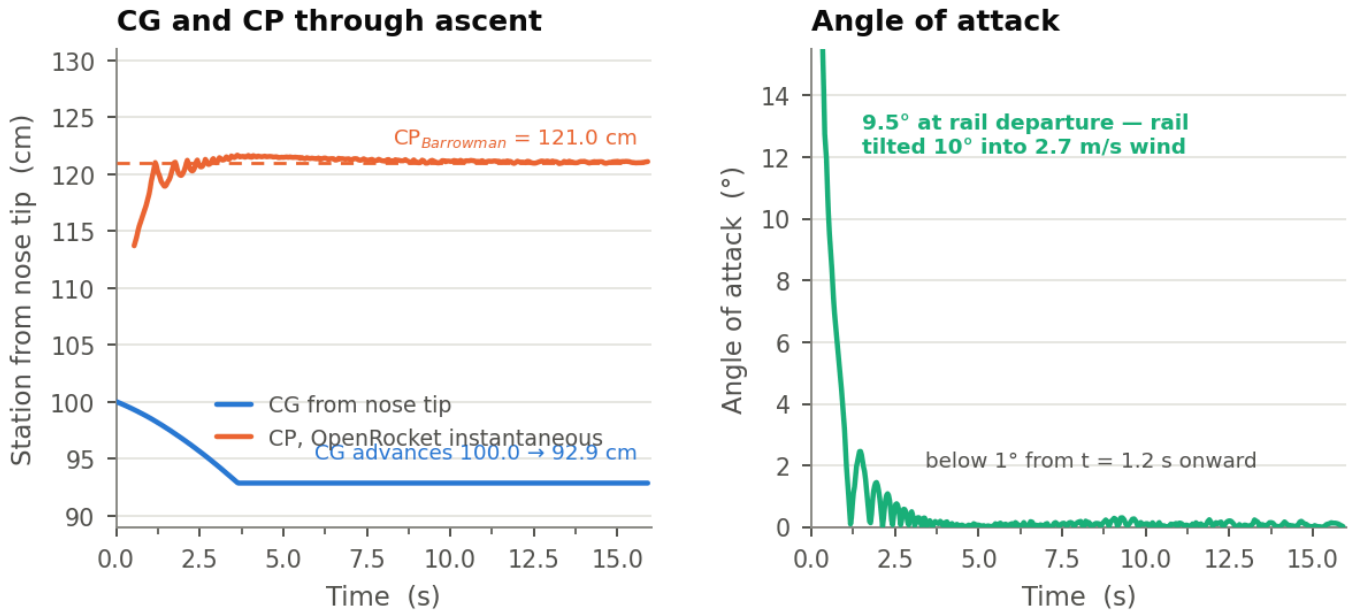


**Figure 7.** Drag coefficient decomposition against Mach number, and the resulting drag force. Skin friction dominates ( $\approx 63\%$  of  $C_D$ ); base drag grows with Mach as expected. The vehicle never approaches the transonic rise.

| Mach   | $C_D$ total | Skin friction | Pressure | Base   | Drag force (N) | V (m/s) |
|--------|-------------|---------------|----------|--------|----------------|---------|
| 0.0560 | 0.4300      | 0.2990        | 0.0110   | 0.1200 | 1.0            | 18.2    |
| 0.1010 | 0.4310      | 0.2980        | 0.0110   | 0.1210 | 2.8            | 33.6    |
| 0.1500 | 0.4330      | 0.2980        | 0.0120   | 0.1230 | 6.2            | 49.8    |
| 0.2010 | 0.4350      | 0.2980        | 0.0130   | 0.1250 | 11.0           | 67.1    |
| 0.2500 | 0.4390      | 0.2970        | 0.0140   | 0.1280 | 17.2           | 83.8    |
| 0.3010 | 0.4440      | 0.2960        | 0.0160   | 0.1320 | 25.1           | 100.9   |
| 0.3490 | 0.4490      | 0.2950        | 0.0180   | 0.1360 | 33.9           | 117.0   |
| 0.4010 | 0.4560      | 0.2940        | 0.0210   | 0.1410 | 45.4           | 134.7   |
| 0.4630 | 0.4660      | 0.2920        | 0.0260   | 0.1480 | 61.2           | 155.3   |

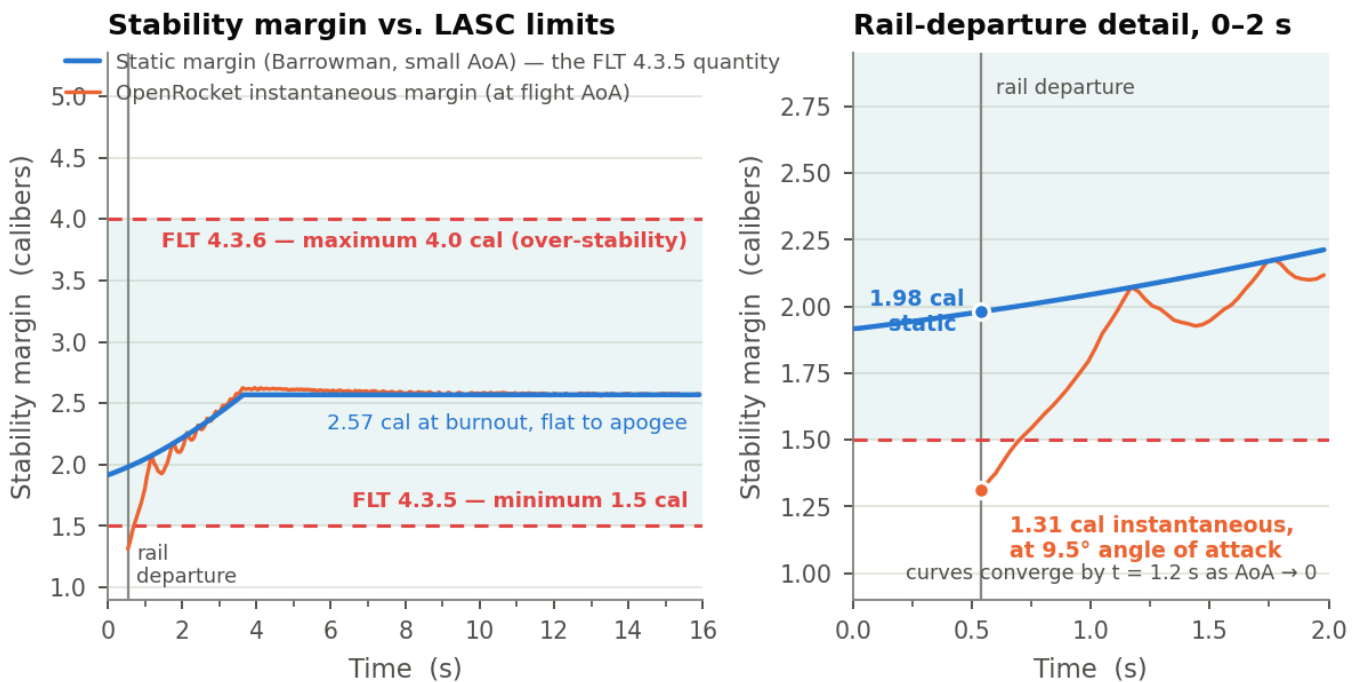
Coefficients are referenced to the 94.34 cm<sup>2</sup> frontal area. Maximum  $C_D$  over the ascent is 0.466 at Mach 0.463; peak drag force is 61.2 N, or 0.59 of the takeoff weight (0.75 of the burnout weight). Peak Mach stays 7.4 % below the 0.5 threshold at which wave-drag effects begin to shift the centre of pressure (RCSM FLT 4.3.5), so the CP solution is valid over the whole ascent.

## 6 STABILITY



**Figure 8.** Centre of gravity and centre of pressure through the ascent, and the angle-of-attack history. The CG advances 7.1 cm forward as propellant is consumed, which increases the margin monotonically.

### 6.1 Static margin against the LASC limits



**Figure 9.** Static stability margin against FLT 4.3.5 ( $\geq 1.5$  cal) and FLT 4.3.6 ( $\leq 4.0$  cal). Blue is the static margin evaluated at small angle of attack — the quantity the rule defines. Orange is OpenRocket’s instantaneous margin, which is evaluated at the flight angle of attack and therefore carries the launch transient.

The static stability margin — CP and CG evaluated in the linear, small-angle-of-attack regime — is **1.92 calibers on the pad, 1.98 calibers at rail departure and 2.57 calibers at burnout**, rising monotonically and remaining inside the [1.5, 4.0] band required by FLT 4.3.5 and FLT 4.3.6 for the entire flight.

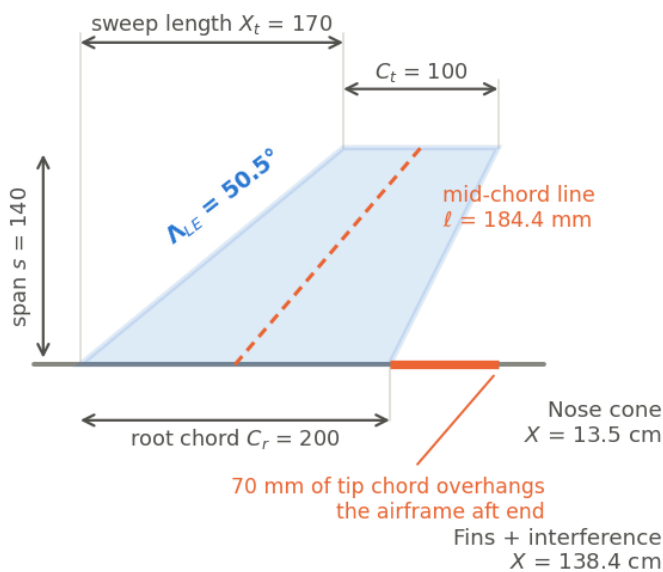
OpenRocket additionally reports an *instantaneous* margin computed at the vehicle’s actual angle of attack. During the first 0.18 s after the vehicle clears the rail this quantity reads 1.31 cal, because the rail is deliberately tilted 10° into a 2.7 m/s wind and the vehicle therefore leaves it at 9.5° angle of attack. The angle of attack decays below 1° by  $t = 1.2$  s (7.4 m AGL) and the two curves converge for the remainder of the flight. The transient is a property of the initial condition, not of the airframe.

### 6.2 Independent analytical verification (Barrowman)

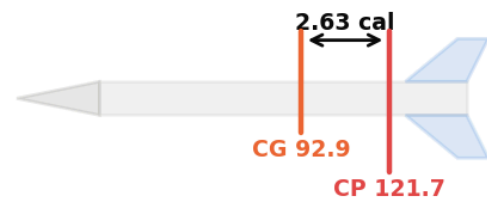
| Term                                            | Value                 |
|-------------------------------------------------|-----------------------|
| Fin mid-chord line $\ell$                       | 184.4 mm              |
| $(C_{N\alpha})_{\text{fins}}$ , 4 fins          | 10.101 /rad           |
| Body-fin interference $K_{fb}$                  | 1.2813                |
| $(C_{N\alpha})_{\text{fins}}$ with interference | 12.942 /rad           |
| Fin CP station $X_f$                            | 138.44 cm             |
| Nose $C_{N\alpha}$ / station                    | 2.000 /rad · 13.50 cm |
| <b>Total <math>C_{N\alpha}</math></b>           | <b>14.942 /rad</b>    |
| <b><math>X_{CP}</math> — hand calculation</b>   | <b>121.72 cm</b>      |
| $X_{CP}$ — OpenRocket                           | 121.03 cm             |
| <b>Agreement</b>                                | <b>0.57 %</b>         |

| Dynamic quantity             | Rail departure       | Max-Q                 |
|------------------------------|----------------------|-----------------------|
| Velocity                     | 16.17 m/s            | 155.33 m/s            |
| Dynamic pressure             | 152.7 Pa             | 13 765 Pa             |
| Corrective moment $C_1$      | 4.83 N·m/rad         | 559.3 N·m/rad         |
| Damping moment $C_2$         | 0.308 N·m·s/rad      | 3.299 N·m·s/rad       |
| Natural frequency $\omega_n$ | 1.73 rad/s (0.28 Hz) | 20.06 rad/s (3.19 Hz) |
| Damping ratio $\zeta$        | 0.055                | 0.059                 |
| Static margin                | 1.98 cal             | 2.63 cal              |

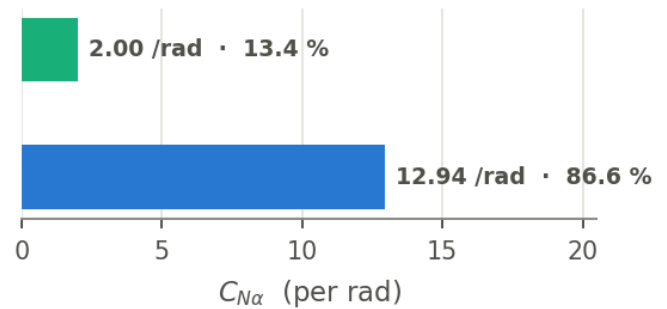
### Fin planform (mm)



### CP and CG at burnout (cm from nose tip)



### Where the stability comes from



**Figure 10.** Fin planform and the origin of the stability solution. The fins and their body interference supply 86.6 % of the vehicle's normal-force slope; the nose supplies the rest.

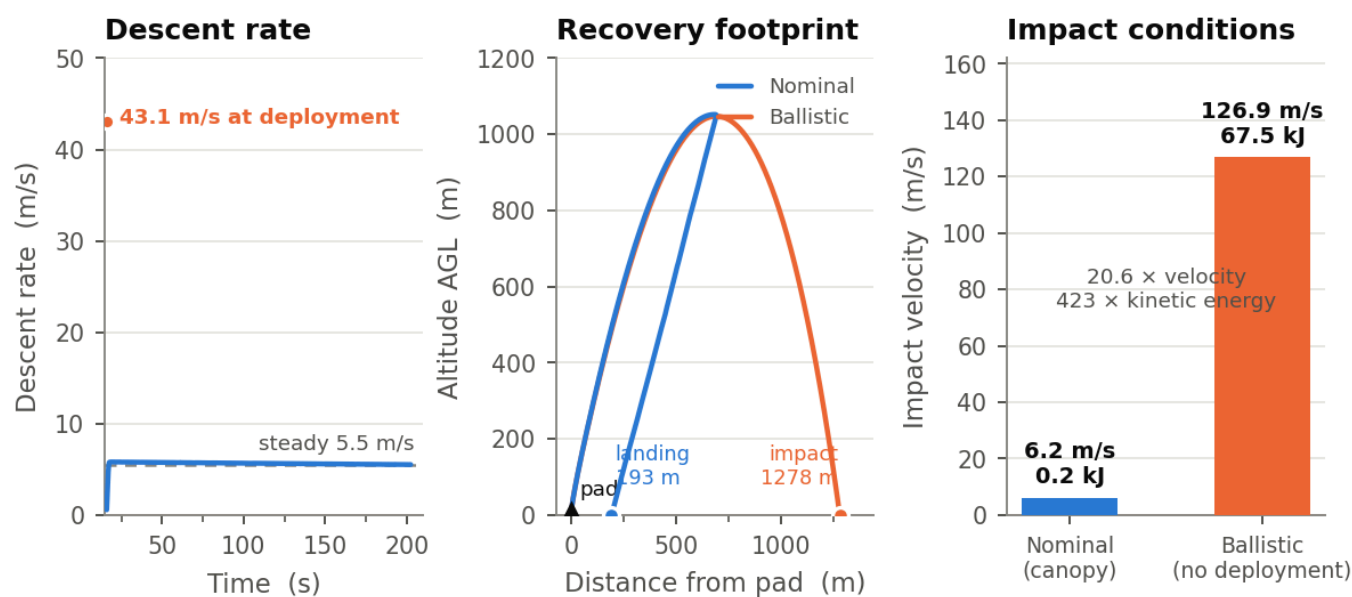
**Why this fin set is the right size.** The four fins place the centre of pressure 121.7 cm aft of the nose tip against a CG that travels from 100.0 to 92.9 cm — a margin of 1.9 to 2.6 calibers. That band is chosen deliberately: below roughly 1.5 calibers the vehicle is sensitive to gusts and to CG uncertainty, while above about 4 calibers it weathercocks strongly into the wind, trading apogee for horizontal drift and enlarging the recovery footprint. At 2.0–2.6 cal the pitch natural frequency at Max-Q is 3.19 Hz with a damping ratio of 0.059, so a disturbance decays with a time constant of 0.85 s — short relative to the 15.9 s ascent, and slow enough that the airframe is never driven at a structurally significant rate. The measured angle-of-attack history confirms it: the 9.5° launch transient is removed inside 0.7 s and the vehicle then flies below 1° for the whole ascent.

### 6.3 Stability data

| t (s) | V (m/s) | Mach  | CG (cm) | CP OR (cm) | Static SM (cal) | Instantaneous SM (cal) | AoA (°) |
|-------|---------|-------|---------|------------|-----------------|------------------------|---------|
| 0.00  | 0.0     | 0.008 | 100.02  | —          | 1.917           | —                      | 80.00   |
| 0.54  | 16.2    | 0.050 | 99.33   | 113.72     | 1.980           | 1.313                  | 9.52    |

| t (s) | V (m/s) | Mach  | CG (cm) | CP OR (cm) | Static SM (cal) | Instantaneous SM (cal) | AoA (°) |
|-------|---------|-------|---------|------------|-----------------|------------------------|---------|
| 0.75  | 23.5    | 0.071 | 99.01   | 115.93     | 2.009           | 1.543                  | 6.26    |
| 0.99  | 32.4    | 0.098 | 98.63   | 118.30     | 2.044           | 1.794                  | 3.20    |
| 1.50  | 53.7    | 0.161 | 97.72   | 119.06     | 2.126           | 1.946                  | 2.35    |
| 2.01  | 76.8    | 0.229 | 96.71   | 120.14     | 2.219           | 2.138                  | 1.20    |
| 2.49  | 99.5    | 0.297 | 95.68   | 120.78     | 2.313           | 2.290                  | 0.60    |
| 3.00  | 124.4   | 0.371 | 94.49   | 121.02     | 2.421           | 2.420                  | 0.50    |
| 3.66  | 155.3   | 0.463 | 92.88   | 121.60     | 2.569           | 2.621                  | 0.11    |
| 6.00  | 121.2   | 0.364 | 92.88   | 121.36     | 2.569           | 2.598                  | 0.12    |
| 10.01 | 76.8    | 0.234 | 92.88   | 121.13     | 2.569           | 2.578                  | 0.12    |
| 15.90 | 43.1    | 0.136 | 92.88   | 121.12     | 2.569           | 2.577                  | 0.02    |

## 7 RECOVERY AND BALLISTIC FOOTPRINT



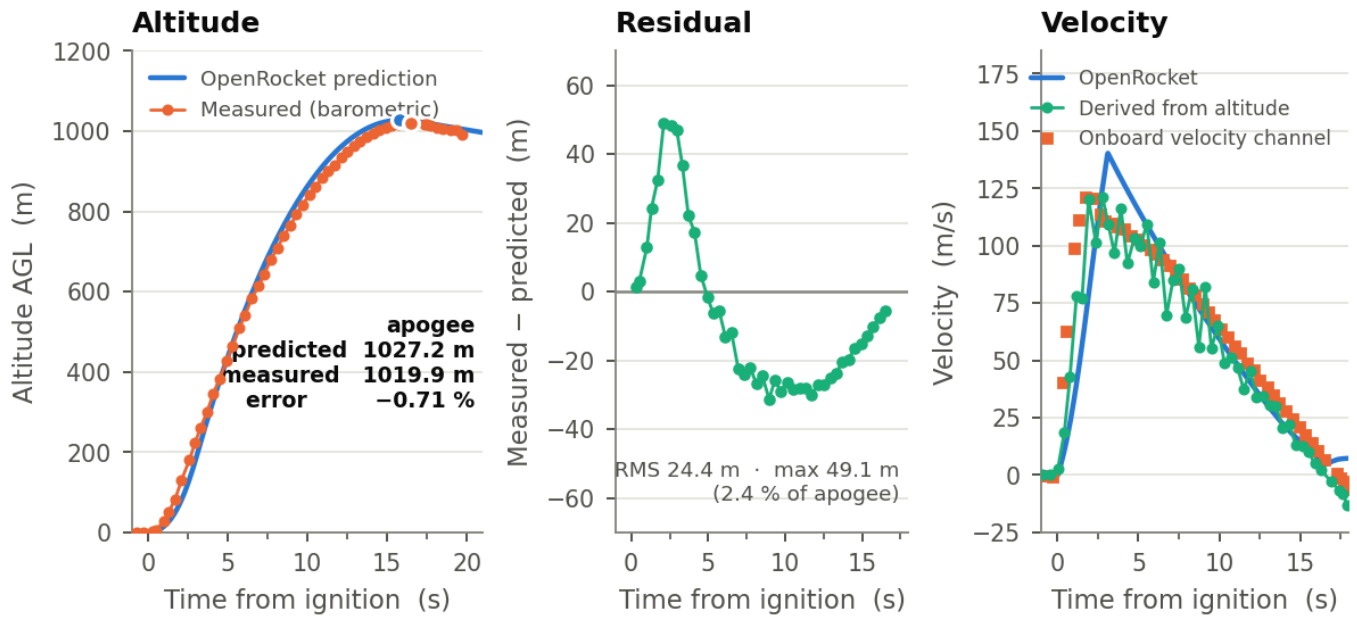
**Figure 11.** Descent rate, recovery footprint for both cases, and impact conditions. The ballistic case is the safety-relevant one: it lands 1278 m downrange at 126.9 m/s.

| Recovery parameter     | Value                             |
|------------------------|-----------------------------------|
| Canopy                 | Flat octagonal, 2.80 m tip-to-tip |
| Drag coefficient       | 0.75                              |
| Shroud lines           | 8 × 1.50 m                        |
| Packed volume          | 210 mm × 100 mm dia.              |
| Deployment             | At apogee, altimeter-triggered    |
| Deployment velocity    | 43.1 m/s                          |
| Steady descent rate    | 5.52 m/s                          |
| Landing kinetic energy | 160 J                             |

| Case      | Impact V   | Energy  | Range  | Ratio              |
|-----------|------------|---------|--------|--------------------|
| Nominal   | 6.17 m/s   | 0.16 kJ | 193 m  | 1×                 |
| Ballistic | 126.86 m/s | 67.5 kJ | 1278 m | <b>423× energy</b> |

The ballistic case is submitted per CRS 10.1.10 and 10.1.11 and is the input to the range-safety assessment. It establishes the downrange keep-out distance (1278 m on the modelled 2.7 m/s wind) and quantifies the consequence of a recovery-system failure, which drives the criticality ranking of the recovery-system failure modes in the mission FMECA. The 43.1 m/s deployment velocity at apogee is a direct consequence of the 10° rail tilt: the residual velocity at apogee is almost entirely horizontal (the vertical component is near zero), so the canopy loading is well inside its rated envelope.

## 8 FLIGHT-TEST CORRELATION — 4 JULY 2026



**Figure 12.** Measured barometric telemetry against the pre-flight OpenRocket prediction for the same vehicle and site. Left: altitude. Centre: residual through the ascent. Right: three independent velocity estimates.

| Quantity                                | Predicted | Measured   | Error              |
|-----------------------------------------|-----------|------------|--------------------|
| Apogee AGL                              | 1027.2 m  | 1019.9 m   | -0.71 %            |
| Time to apogee                          | 15.79 s   | 16.52 s    | +4.6 %             |
| Rail departure velocity<br>(3.0 m rail) | 13.16 m/s | ≈ 18.5 m/s | +41 %              |
| Peak velocity (onboard<br>channel)      | 140.3 m/s | 121.4 m/s  | -13.5 %            |
| Ascent altitude RMS<br>residual         | —         | 24.4 m     | 2.4 % of<br>apogee |

The vehicle flown on 4 July 2026 used the same airframe and recovery architecture at a higher-elevation site (Pachuca, 2380 m MSL). Apogee — the single quantity the competition scores — was predicted to within 0.71 %. The ascent profile tracks the prediction with a 24.4 m RMS residual over 42 telemetry packets, or 2.4 % of apogee.

Rail departure was measured empirically: the barometric channel records the vehicle passing from 2.4 m to 6.1 m in 200 ms, an average of 18.5 m/s across the 3.0 m rail exit, against 13.2 m/s predicted. The model is therefore **conservative** on rail exit velocity, which is the direction that matters for FLT 4.3.4.

| Packet | t (s)  | Measured (m) | Predicted (m) | Δ (m) | Vel. channel (m/s) |
|--------|--------|--------------|---------------|-------|--------------------|
| 77     | 0.318  | 2.4          | 1.1           | +1.3  | 40.3               |
| 78     | 0.518  | 6.1          | 3.2           | +2.9  | 62.8               |
| 80     | 1.318  | 50.9         | 26.6          | +24.3 | 111.2              |
| 83     | 2.619  | 180.7        | 132.4         | +48.3 | 113.7              |
| 86     | 3.718  | 299.8        | 277.8         | +22.0 | 108.1              |
| 89     | 4.960  | 428.8        | 430.5         | -1.7  | 103.0              |
| 92     | 6.118  | 542.1        | 555.3         | -13.2 | 96.6               |
| 95     | 7.318  | 643.9        | 667.9         | -24.0 | 88.0               |
| 98     | 8.518  | 739.7        | 764.2         | -24.5 | 78.6               |
| 101    | 9.718  | 815.9        | 844.8         | -28.9 | 67.5               |
| 104    | 10.960 | 884.2        | 912.3         | -28.1 | 56.3               |
| 107    | 12.118 | 934.0        | 961.0         | -27.0 | 45.6               |
| 110    | 13.318 | 973.6        | 997.2         | -23.6 | 34.7               |
| 113    | 14.518 | 1002.7       | 1019.2        | -16.5 | 24.4               |
| 116    | 15.718 | 1017.1       | 1027.2        | -10.1 | 14.1               |
| 118    | 16.518 | 1019.9       | 1025.5        | -5.6  | 6.9                |

The onboard velocity channel is a derived quantity and is included for completeness. Over the coast phase it agrees with the altitude-derived velocity to within about 10 %. In the first two packets after motor ignition it reads high (40–63 m/s at 2.4–6.1 m AGL) relative to both the measured displacement and the thrust available from the motor; the barometric altitude channel — which is the one the apogee determination and this correlation rest on — is unaffected. Improving the velocity estimator through the ignition transient is a tracked avionics action for the next flight.

## 9 REQUIREMENTS COMPLIANCE

| Requirement           | Criterion                                                                 | This mission                                                                                             | Status           |
|-----------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------|
| CRS 10.1.3 / 10.1.7   | Runnable simulation file, OpenRocket or RocketPy                          | .ork + .eng delivered in .zip                                                                            | Met              |
| CRS 10.1.8            | Exact masses, sizes, relative positions modelled                          | Component-level model, budget closes on 10.520 kg                                                        | Met              |
| CRS 10.1.9            | .eng supplied for SRAD motor; recovery modelled with trigger, size, $C_D$ | Icarus-I_K503.eng ; canopy 2.80 m, $C_D$ 0.75, apogee trigger                                            | Met              |
| CRS 10.1.10 / 10.1.11 | Ballistic <i>and</i> nominal cases submitted                              | Both delivered                                                                                           | Met              |
| FLT 4.3.1             | Elevation $80^\circ \pm 5^\circ$                                          | $80.0^\circ$ modelled                                                                                    | Met              |
| FLT 4.3.3             | Stable in gusts to 15 m/s                                                 | Modelled at 2.7 m/s + 10 % turbulence; margin $\geq 1.9$ cal                                             | Met              |
| FLT 4.3.4             | Rail departure $\geq 30$ m/s, <i>or</i> detailed analysis above 15 m/s    | 16.17 m/s simulated on the 4.0 m rail, supported by \$6 analysis and by 18.5 m/s measured in flight test | Met via analysis |
| FLT 4.3.5             | Static margin $\geq 1.5$ cal throughout ascent                            | 1.92 $\rightarrow$ 2.57 cal; minimum 1.92 cal on the pad                                                 | Met              |
| FLT 4.3.6             | Static margin $\leq 4.0$ cal                                              | Maximum 2.57 cal                                                                                         | Met              |
| FLT 4.4.1             | Prepared for the 4 m LASC rail                                            | 4.0 m rail modelled                                                                                      | Met              |
| PRS 5.1.3             | Installed total impulse $\leq 40\,960$ N·s                                | 1855.9 N·s                                                                                               | Met              |
| PRS 5.1.4             | Thrust-to-weight $\geq 5:1$ (greater of initial or average thrust)        | 514.1 N $\div$ 103.2 N = <b>4.98:1</b> at the modelled 10.520 kg                                         | Verify at LRR    |

**PRS 5.1.4 — open item.** With the average thrust of 514.1 N and the modelled takeoff mass of 10.520 kg the thrust-to-weight ratio is 4.98:1, against a 5:1 minimum (standard gravity). The requirement is satisfied at any takeoff mass of **10.48 kg or less**. The as-built vehicle is to be weighed in flight configuration at the Launch Readiness Review and the ratio recomputed on the measured mass; a 40 g reduction clears the requirement with margin. Every other trajectory result in this report is insensitive to a change of this size — a 40 g reduction raises predicted apogee by less than 10 m.

## 10 FILES DELIVERED

| File                                      | Content                                              |
|-------------------------------------------|------------------------------------------------------|
| MissionPROMETEO_Nominal_RocketPy_v1.ork   | OpenRocket model, nominal flight case active         |
| MissionPROMETEO_Ballistic_RocketPy_v1.ork | Same model, recovery inhibited                       |
| Icarus-I_K503.eng                         | SRAD motor thrust curve (CRS 10.1.9)                 |
| PrometeoStellaIgnis2026.csv               | Full nominal time history, 58 variables, 1088 points |
| motorprometeo.csv                         | OpenMotor internal-ballistics output                 |
| simulation_data_appendix.xlsx             | All tables in this report, machine-readable          |

Naming per CRS 10.1.6. All results in this report were produced with OpenRocket 24.12 from the delivered model files and are reproducible by a LASC or RocketPy team member from those files alone (CRS 10.1.5). Per CRS 10.2.1, the simulation inputs will be re-verified against the as-measured vehicle data recorded at the Launch Readiness Review, and the trajectory re-run on site if the launch elevation is adjusted by event officials (FLT 4.3.2).